Homework 4: Solutions

STT 861 — Theory of Probability and Statistics I

Casella and Berger, *Statistical Inference*, Chapter 3, pp. 127–131

Homework problems: 1–10, corresponding to Exercises 3.1, 3.4, 3.5, 3.6, 3.8, 3.10, 3.11, 3.13, 3.20, and 3.24.

Problem 1 — C&B Exercise 3.1, p. 127

Find expressions for $\mathrm{E}X$ and $\mathrm{Var}\,X$ if X is a random variable with the general discrete uniform(N_0, N_1) distribution that puts equal probability on each of the values $N_0, N_0+1, \ldots, N_1$. Here $N_0 \le N_1$ and both are integers.

Solution

Put $m = N_1 - N_0 + 1$ and $U = X - N_0$. Then $\mathbb{P}(U = u) = 1/m$ for $u = 0, \ldots, m-1$. The formulas for sums of integers and squares give

$$\mathbb{E}U = \frac{1}{m}\sum_{u=0}^{m-1} u = \frac{m-1}{2}, \qquad \mathbb{E}U^2 = \frac{1}{m}\sum_{u=0}^{m-1} u^2 = \frac{(m-1)(2m-1)}{6}.$$

Consequently,

$$\boxed{\mathbb{E}X = N_0 + \frac{m-1}{2} = \frac{N_0+N_1}{2}},$$

and, since a location shift does not change variance,

$$\begin{aligned}\mathrm{Var}\,X = \mathrm{Var}\,U &= \frac{(m-1)(2m-1)}{6} - \frac{(m-1)^2}{4} \\ &= \boxed{\frac{m^2-1}{12} = \frac{(N_1-N_0)(N_1-N_0+2)}{12}}.\end{aligned}$$

These formulas include the case $N_0 = N_1$, when X is constant and its variance is zero.

Problem 2 — C&B Exercise 3.4, p. 128

A man with n keys wants to open his door and tries the keys at random. Exactly one key will open the door. Find the mean number of trials if

(a) unsuccessful keys are not eliminated from further selections.

(b) unsuccessful keys are eliminated.

Solution

Let T count all trials up to and including the successful trial.

(a) Unsuccessful keys are retained. Each trial independently selects one of the n keys uniformly, so T is geometric with success probability $1/n$:

$$\mathbb{P}(T = t) = \left(1 - \frac{1}{n}\right)^{t-1}\frac{1}{n}, \qquad t = 1, 2, \ldots.$$

Using the tail-sum formula for a positive integer-valued random variable,

$$\boxed{\mathbb{E}T = \sum_{t=0}^{\infty} \mathbb{P}(T > t) = \sum_{t=0}^{\infty} \left(1 - \frac{1}{n}\right)^t = n}.$$

(b) Unsuccessful keys are removed. Trying the remaining keys uniformly at each step is equivalent to trying all keys in a uniformly random order. The successful key has an equally likely position in $\{1, \ldots, n\}$. Hence

$$\mathbb{P}(T = t) = \frac{1}{n}, \quad t = 1, \ldots, n, \qquad \boxed{\mathbb{E}T = \frac{1}{n} \sum_{t=1}^{n} t = \frac{n+1}{2}}.$$

Problem 3 C&B Exercise 3.5, p. 128

A standard drug is known to be effective in 80% of the cases in which it is used. A new drug is tested on 100 patients and found to be effective in 85 cases. Is the new drug superior? (*Hint:* Evaluate the probability of observing 85 or more successes assuming that the new and old drugs are equally effective.)

Solution

Assuming independent patient responses with a common success probability p, let X be the number of successes. Under the hypothesis that the new drug is equally effective, $X \sim \text{Binomial}(100, 0.8)$. The exact upper-tail probability is

$$\boxed{\mathbb{P}_{p=0.8}(X \geq 85) = \sum_{k=85}^{100} \binom{100}{k} (0.8)^k (0.2)^{100-k} \approx 0.1285}.$$

This is the one-sided p-value for testing equal efficacy against $p > 0.8$. Thus an observed success rate of 85% does not provide strong evidence of superiority: the one-sided test does not reject at the conventional 5% level. The observed improvement can reasonably be explained by sampling variation.

Problem 4 C&B Exercise 3.6, p. 128

A large number of insects are expected to be attracted to a certain variety of rose plant. A commercial insecticide is advertised as being 99% effective. Suppose 2,000 insects infest a rose garden where the insecticide has been applied, and let $X =$ number of surviving insects.

(a) What probability distribution might provide a reasonable model for this experiment?

(b) Write down, but do not evaluate, an expression for the probability that fewer than 100 insects survive, using the model in part (a).

(c) Evaluate an approximation to the probability in part (b).

Solution

(a) Model. Treat survival of each insect as an independent Bernoulli trial with survival probability 0.01. Then

$$\boxed{X \sim \text{Binomial}(2000, 0.01)}.$$

In particular, $\mathbb{E}X = 20$ and $\operatorname{Var} X = 2000(0.01)(0.99) = 19.8$.

(b) Exact expression. Fewer than 100 means $X \le 99$, so

$$\boxed{\mathbb{P}(X < 100) = \sum_{x=0}^{99} \binom{2000}{x} (0.01)^x (0.99)^{2000-x}}.$$

(c) Poisson approximation. Since the number of trials is large and the survival probability is small, use $Y \sim \text{Poisson}(2000 \cdot 0.01) = \text{Poisson}(20)$. Thus

$$\boxed{\mathbb{P}(X < 100) \approx \mathbb{P}(Y \le 99) = e^{-20} \sum_{x=0}^{99} \frac{20^x}{x!} \approx 1 - 3.4889 \times 10^{-37} \approx 1}.$$

For a stable numerical evaluation of the tiny complementary probability, one may sum

$$\mathbb{P}(Y \ge 100) = e^{-20} \frac{20^{100}}{100!} \left(1 + \frac{20}{101} + \frac{20^2}{101 \cdot 102} + \cdots\right).$$

The displayed complement is for the Poisson approximation; the probability of 100 or more survivors is not exactly zero.

Problem 5 — C&B Exercise 3.8, p. 128

Two movie theaters compete for the business of 1,000 customers. Assume that each customer chooses between the movie theaters independently and with "indifference." Let N denote the number of seats in each theater.

(a) Using a binomial model, find an expression for N that will guarantee that the probability of turning away a customer (because of a full house) is less than 1%.

(b) Use the normal approximation to get a numerical value for N.

Solution

(a) Exact binomial condition. Let X count customers choosing theater 1. Then X is binomial with parameters 1000 and 1/2, and theater 2 receives $1000 - X$ customers. No customer is turned away exactly when

$$X \le N \quad \text{and} \quad 1000 - X \le N, \qquad \text{or equivalently} \qquad 1000 - N \le X \le N.$$

If $N < 500$, turning away customers is unavoidable. For $500 \le N \le 1000$, the two overflow events $X > N$ and $X < 1000 - N$ are disjoint and have the same probability by binomial symmetry. Therefore the required condition is

$$\boxed{2 \sum_{x=N+1}^{1000} \binom{1000}{x} 2^{-1000} < 0.01}.$$

Equivalently, choose the smallest integer $N \ge 500$ for which $\mathbb{P}(X \le N) > 0.995$.

(b) Normal approximation with continuity correction. The binomial mean is 500 and the variance is 250. If Φ is the standard normal cdf, then

$$\mathbb{P}(\text{at least one customer turned away}) \approx 2\left[1 - \Phi\left(\frac{N + 0.5 - 500}{\sqrt{250}}\right)\right].$$

To make this less than 0.01, require

$$\frac{N + 0.5 - 500}{\sqrt{250}} > z_{0.995} \approx 2.575829, \qquad N > 499.5 + 2.575829\sqrt{250} \approx 540.2274.$$

Thus the normal approximation gives

$$\boxed{N = 541 \text{ seats in each theater}}.$$

As a check, the exact overflow probabilities are approximately 0.0103881 for $N = 540$ and 0.00863995 for $N = 541$. Since this probability decreases as N increases, 541 is also the smallest capacity satisfying the exact binomial condition.

Problem 6 — C&B Exercise 3.10, p. 129

Shuster (1991) describes a number of probability calculations that he did for a court case involving the sale of cocaine. A Florida police department had seized 496 *suspected* packets of cocaine, of which four were randomly selected and tested and found to actually be cocaine. The police then chose two more packets at random and, posing as drug dealers, sold the packets to the defendant. These last two packets were lost before they could be tested to verify that they were, indeed, cocaine.

(a) If the original 496 packets were composed of N packets of cocaine and $M = 496 - N$ noncocaine, show that the probability of selecting 4 cocaine packets and then 2 noncocaine packets, which is the probability that the defendant is innocent of buying cocaine, is

$$\frac{\binom{N}{4}\binom{M}{2}}{\binom{N+M}{4}\binom{N+M-4}{2}}.$$

(b) Maximizing (in M and N) the probability in part (a) maximizes the defendant's "innocence probability". Show that this probability is .022, attained at $M = 165$ and $N = 331$.

Solution

(a) The sampling probability. The probability that the first sample consists of four cocaine packets is $\binom{N}{4}/\binom{496}{4}$. Conditional on this event, 492 packets remain, including all M noncocaine packets. Thus

$$\boxed{q(N, M) = \frac{\binom{N}{4}\binom{M}{2}}{\binom{N+M}{4}\binom{N+M-4}{2}}}, \qquad N + M = 496.$$

This is the joint probability of the two specified sampling outcomes for a fixed composition (N, M).

(b) Maximize over the integer compositions. The denominator is constant, so maximize $h(N) = \binom{N}{4}\binom{496-N}{2}$. Its positive values occur for $4 \le N \le 494$. For $4 \le N \le 493$,

$$\frac{h(N+1)}{h(N)} = \frac{(N+1)(494-N)}{(N-3)(496-N)}.$$

The denominator is positive, and the difference between the numerator and denominator is

$$(N+1)(494-N) - (N-3)(496-N) = 1982 - 6N.$$

Consequently $h(N+1) > h(N)$ for integers $N \le 330$, and $h(N+1) < h(N)$ for integers $N \ge 331$. The unique maximum is therefore attained at $N = 331$, $M = 165$, and

$$\boxed{\max q(N, M) = \frac{\binom{331}{4}\binom{165}{2}}{\binom{496}{4}\binom{492}{2}} = 0.02208168087\ldots \approx 0.022.}$$

Problem 7 **C&B Exercise 3.11, p. 129**

The hypergeometric distribution can be approximated by either the binomial or the Poisson distribution. (Of course, it can be approximated by other distributions, but in this exercise we will concentrate on only these two.) Let X have the hypergeometric distribution

$$P(X = x \mid N, M, K) = \frac{\binom{M}{x}\binom{N-M}{K-x}}{\binom{N}{K}}, \qquad x = 0, 1, \ldots, K.$$

(a) Show that as $N \to \infty$, $M \to \infty$, and $M/N \to p$,

$$P(X = x \mid N, M, K) \to \binom{K}{x} p^x (1-p)^{K-x}, \qquad x = 0, 1, \ldots, K.$$

(Stirling's Formula from Exercise 1.23 may be helpful.)

(b) Use the fact that the binomial can be approximated by the Poisson to show that if $N \to \infty$, $M \to \infty$, $K \to \infty$, $M/N \to 0$, and $KM/N \to \lambda$, then

$$P(X = x \mid N, M, K) \to \frac{e^{-\lambda}\lambda^x}{x!}, \qquad x = 0, 1, \ldots.$$

(c) Verify the approximation in part (b) directly, without using the Poisson approximation to the binomial. (Lemma 2.3.14 is helpful.)

Solution

An empty product equals one.

(a) Binomial limit with fixed sample size.

Step 1: Write the pmf using ordinary factorials. First suppose $0 < p < 1$. For fixed K and $0 \le x \le K$,

$$\mathbb{P}(X = x) = \binom{K}{x} \frac{M!}{(M-x)!} \frac{(N-M)!}{(N-M-K+x)!} \frac{(N-K)!}{N!}.$$

Step 2: Apply Stirling's Formula to the factorial ratios. Stirling's Formula is

$$L! \sim \sqrt{2\pi L}\left(\frac{L}{e}\right)^L \qquad (L \to \infty).$$

For every fixed nonnegative integer t, applying this formula to both factorials gives

$$\begin{aligned}\frac{L!}{(L-t)!} &\sim \frac{\sqrt{2\pi L}(L/e)^L}{\sqrt{2\pi(L-t)}((L-t)/e)^{L-t}} \\ &= L^t e^{-t}\left(1 - \frac{t}{L}\right)^{-L+t-1/2} \sim L^t.\end{aligned}$$

The final asymptotic follows because $(1 - t/L)^L \to e^{-t}$, an application of Lemma 2.3.14 with the constant sequence $a_L = -t$, while $(1 - t/L)^{t-1/2} \to 1$. Thus *Stirling's Formula is used here to replace each factorial ratio by its leading power.* Since $M \to \infty$ and $N - M \to \infty$,

$$\begin{aligned}\frac{M!}{(M-x)!} &\sim M^x, \\ \frac{(N-M)!}{(N-M-K+x)!} &\sim (N-M)^{K-x}, \\ \frac{(N-K)!}{N!} &\sim N^{-K}.\end{aligned}$$

Step 3: Take the limit. Substitution in Step 1 gives

$$\mathbb{P}(X = x) \sim \binom{K}{x} \left(\frac{M}{N}\right)^x \left(1 - \frac{M}{N}\right)^{K-x}.$$

Therefore

$$\boxed{\mathbb{P}(X = x) \longrightarrow \binom{K}{x} p^x (1-p)^{K-x}, \qquad x = 0, \ldots, K.}$$

For $p = 0$ or $p = 1$, use the equivalent finite-product identity

$$\mathbb{P}(X = x) = \binom{K}{x} \frac{\displaystyle\prod_{j=0}^{x-1} \left(\frac{M}{N} - \frac{j}{N}\right) \prod_{j=0}^{K-x-1} \left(1 - \frac{M}{N} - \frac{j}{N}\right)}{\displaystyle\prod_{j=0}^{K-1} \left(1 - \frac{j}{N}\right)}.$$

There are only finitely many factors, so taking their limits directly gives the same binomial limit, degenerate at 0 or K, respectively.

(b) Poisson limit through a binomial approximation. Set $p_N = M/N$. The assumptions give

$$p_N \to 0, \qquad K p_N \to \lambda, \qquad \frac{K}{N} = \frac{K p_N}{M} \to 0.$$

Because K now increases, we first justify the binomial approximation in this setting.

Let $U_1, \ldots, U_K$ be independent uniform$(0, 1)$ variables. Construct the indicators of success in sampling without replacement recursively:

$$S_0 = 0, \qquad q_i = \frac{M - S_{i-1}}{N - i + 1}, \qquad I_i = \mathbf{1}_{\{U_i \le q_i\}}, \qquad S_i = S_{i-1} + I_i.$$

Then S_K has the hypergeometric distribution of X. Using the same uniforms, define $Z_i = \mathbf{1}_{\{U_i \le p_N\}}$ and $Y = \sum_{i=1}^K Z_i$. Thus $Y \sim$ Binomial(K, p_N). Since $\mathbb{E}S_{i-1} = (i-1)p_N$,

$$\begin{aligned}
\mathbb{P}(S_K \neq Y) &\le \sum_{i=1}^K \mathbb{P}(I_i \neq Z_i) = \sum_{i=1}^K \mathbb{E}|q_i - p_N| \\
&= \sum_{i=1}^K \frac{\mathbb{E}|(i-1)p_N - S_{i-1}|}{N - i + 1} \\
&\le \sum_{i=1}^K \frac{2(i-1)p_N}{N - i + 1} \le \frac{p_N K(K-1)}{N - K + 1} \longrightarrow 0.
\end{aligned}$$

The last limit follows from boundedness of Kp_N and $K/N \to 0$. Hence, for every fixed x,

$$|\mathbb{P}(X = x) - \mathbb{P}(Y = x)| \le \mathbb{P}(S_K \neq Y) \longrightarrow 0.$$

Application of Lemma 2.3.14 in the binomial-to-Poisson step. The lemma states that if $a_n \to a$, then $(1 + a_n/n)^n \to e^a$. Here the exponent is $K \to \infty$ and $-Kp_N \to -\lambda$, so

$$(1 - p_N)^K = \left(1 + \frac{-Kp_N}{K}\right)^K \longrightarrow e^{-\lambda}.$$

For each fixed nonnegative integer x,

$$\begin{aligned}\mathbb{P}(Y=x) &= \binom{K}{x} p_N^x (1-p_N)^{K-x} \\ &= \frac{1}{x!}\left[\prod_{j=0}^{x-1}\left[(K-j)p_N\right]\right](1-p_N)^K(1-p_N)^{-x} \\ &\longrightarrow \frac{\lambda^x}{x!}e^{-\lambda}.\end{aligned}$$

In the product, $(K-j)p_N = Kp_N - jp_N \to \lambda$ for every fixed j. Combining this binomial limit with the coupling bound gives

$$\boxed{\mathbb{P}(X=x) \longrightarrow e^{-\lambda}\frac{\lambda^x}{x!}, \qquad x=0,1,2,\ldots.}$$

If $\lambda = 0$, the limit is the point mass at zero; this also follows immediately from $\mathbb{P}(X \geq 1) \leq \mathbb{E}X = Kp_N \to 0$.

(c) Direct verification from the hypergeometric pmf. The given assumptions imply $K/N = (KM/N)/M \to 0$. Hence $M + K < N$ for all sufficiently large N.

Step 1: Bound the probability of zero successes. From the pmf,

$$\mathbb{P}(X=0) = \frac{\binom{N-M}{K}}{\binom{N}{K}} = \prod_{j=0}^{K-1}\left(1 - \frac{M}{N-j}\right).$$

Each factor lies between $1 - M/(N-K+1)$ and $1 - M/N$. All these numbers are eventually positive, so

$$\left(1 - \frac{M}{N-K+1}\right)^K \leq \mathbb{P}(X=0) \leq \left(1 - \frac{M}{N}\right)^K.$$

Step 2: Apply Lemma 2.3.14 to both bounds. For the upper bound, $-KM/N \to -\lambda$, and therefore the lemma gives

$$\left(1 - \frac{M}{N}\right)^K = \left(1 + \frac{-KM/N}{K}\right)^K \longrightarrow e^{-\lambda}.$$

For the lower bound,

$$-\frac{KM}{N-K+1} = -\frac{KM}{N}\frac{N}{N-K+1} \longrightarrow -\lambda,$$

since $K/N \to 0$. Applying the same lemma again gives

$$\left(1 - \frac{M}{N-K+1}\right)^K = \left(1 + \frac{-KM/(N-K+1)}{K}\right)^K \longrightarrow e^{-\lambda}.$$

These are the applications of Lemma 2.3.14 in the direct proof. The squeeze theorem now yields

$$\mathbb{P}(X=0) \longrightarrow e^{-\lambda}.$$

Step 3: Recover each fixed positive count using explicit products. For a fixed integer $x \geq 1$,

$$\begin{aligned}\frac{\mathbb{P}(X=x)}{\mathbb{P}(X=0)} &= \binom{M}{x}\frac{\binom{N-M}{K-x}}{\binom{N-M}{K}} \\ &= \frac{[M(M-1)\cdots(M-x+1)][K(K-1)\cdots(K-x+1)]}{x!\,(N-M-K+1)(N-M-K+2)\cdots(N-M-K+x)} \\ &= \frac{1}{x!}\left(\frac{MK}{N}\right)^x \prod_{j=0}^{x-1}\frac{(1-j/M)(1-j/K)}{1-(M+K-j-1)/N} \\ &\longrightarrow \frac{\lambda^x}{x!}.\end{aligned}$$

Here $M, K \to \infty$, $M/N \to 0$, and $K/N \to 0$, so every factor in the last finite product tends to one. Multiplying by the limit of $\mathbb{P}(X=0)$ proves

$$\boxed{\mathbb{P}(X=x) \longrightarrow e^{-\lambda}\frac{\lambda^x}{x!}, \qquad x=0,1,2,\ldots.}$$

This proof uses Lemma 2.3.14 directly on the hypergeometric zero probability, without using a binomial-to-Poisson approximation.

Problem 8 C&B Exercise 3.13, p. 130

A *truncated discrete distribution* is one in which a particular class cannot be observed and is eliminated from the sample space. In particular, if X has range $0, 1, 2, \ldots$ and the 0 class cannot be observed (as is usually the case), the 0-truncated random variable X_T has pmf

$$P(X_T = x) = \frac{P(X=x)}{P(X>0)}, \qquad x = 1, 2, \ldots.$$

Find the pmf, mean, and variance of the 0-truncated random variable starting from

(a) $X \sim \text{Poisson}(\lambda)$.

(b) $X \sim$ negative binomial(r, p), as in (3.2.10).

Solution

Zero truncation means conditioning on $X > 0$. Write $d = \mathbb{P}(X > 0) > 0$ and let X_T have the conditional distribution of X given $X > 0$. For $m = 1, 2$, the term at $X = 0$ contributes zero, so

$$\mathbb{E}(X_T^m) = \frac{\mathbb{E}(X^m)}{d}.$$

Consequently, if $\mu = \mathbb{E}X$ and $\sigma^2 = \operatorname{Var} X$,

$$\mathbb{E}X_T = \frac{\mu}{d}, \qquad \operatorname{Var} X_T = \frac{\sigma^2+\mu^2}{d} - \frac{\mu^2}{d^2} = \frac{\sigma^2}{d} - \frac{\mu^2(1-d)}{d^2}.$$

(a) Zero-truncated Poisson. For $\lambda > 0$, $\mathbb{P}(X=0) = e^{-\lambda}$, $\mathbb{E}X = \lambda$, and $\operatorname{Var} X = \lambda$. Hence

$$\boxed{\mathbb{P}(X_T = x) = \frac{e^{-\lambda}\lambda^x}{x!(1-e^{-\lambda})} = \frac{\lambda^x}{x!(e^\lambda - 1)}, \qquad x = 1, 2, \ldots.}$$

The probabilities sum to one because the original positive-class probabilities sum to $1 - e^{-\lambda}$. The moment formulas give

$$\boxed{\mathbb{E}X_T = \frac{\lambda}{1-e^{-\lambda}}, \qquad \operatorname{Var} X_T = \frac{\lambda}{1-e^{-\lambda}} - \frac{\lambda^2 e^{-\lambda}}{(1-e^{-\lambda})^2}.}$$

(b) Zero-truncated negative binomial. For integer r, the book's negative binomial variable counts failures before r successes. Its pmf, also valid for real $r > 0$, is

$$\mathbb{P}(X = x) = \frac{\Gamma(r+x)}{\Gamma(r)x!}\, p^r (1-p)^x, \qquad x = 0, 1, \ldots, \quad r > 0, \quad 0 < p < 1.$$

For integer r the coefficient is $\binom{r+x-1}{x}$. In this parameterization,

$$\mathbb{P}(X = 0) = p^r, \qquad \mu = \frac{r(1-p)}{p}, \qquad \sigma^2 = \frac{r(1-p)}{p^2}.$$

Therefore

$$\boxed{\mathbb{P}(X_T = x) = \frac{\Gamma(r+x)}{\Gamma(r)x!}\, \frac{p^r(1-p)^x}{1-p^r}, \qquad x = 1, 2, \ldots,}$$

and

$$\boxed{\mathbb{E}X_T = \frac{r(1-p)}{p(1-p^r)}.}$$

For the variance,

$$\begin{aligned} \operatorname{Var} X_T &= \frac{r(1-p)}{p^2(1-p^r)} - \frac{r^2(1-p)^2 p^r}{p^2(1-p^r)^2} \\ &= \boxed{\frac{r(1-p)\left[1-p^r - r(1-p)p^r\right]}{p^2(1-p^r)^2}}. \end{aligned}$$

The original distribution has positive mass $1 - p^r$ on the retained classes, which verifies normalization of the truncated pmf.

Problem 9 — C&B Exercise 3.20, p. 131

Let the random variable X have the pdf

$$f(x) = \frac{2}{\sqrt{2\pi}} e^{-x^2/2}, \qquad 0 < x < \infty.$$

(a) Find the mean and variance of X. (This distribution is sometimes called a *folded normal*.)

(b) If X has the folded normal distribution, find the transformation $g(X) = Y$ and values of α and β so that $Y \sim \text{gamma}(\alpha, \beta)$.

Solution

(a) Mean and variance. The density is twice the positive half of the standard normal density. Direct integration, using $u = x^2/2$, yields

$$\mathbb{E}X = \sqrt{\frac{2}{\pi}} \int_0^\infty x e^{-x^2/2}\, dx = \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-u}\, du = \boxed{\sqrt{\frac{2}{\pi}}}.$$

By symmetry of the standard normal density, its second moment on $(0, \infty)$ is half of its full second moment. Thus

$$\mathbb{E}X^2 = 2\int_0^\infty x^2 \frac{e^{-x^2/2}}{\sqrt{2\pi}}\,dx = 1, \qquad \boxed{\operatorname{Var} X = 1 - \frac{2}{\pi}}.$$

(b) Gamma transformation. Set $Y = g(X) = X^2$. Since $x = \sqrt{y}$ and $dx/dy = 1/(2\sqrt{y})$,

$$f_Y(y) = \frac{2}{\sqrt{2\pi}} e^{-y/2} \frac{1}{2\sqrt{y}} = \frac{y^{1/2-1} e^{-y/2}}{\Gamma(1/2)\, 2^{1/2}}, \qquad y > 0.$$

This is the gamma density with shape $\alpha = 1/2$ and scale $\beta = 2$:

$$\boxed{Y = X^2 \sim \text{Gamma}\left(\frac{1}{2}, 2\right).}$$

Equivalently, $X^2/2$ has shape $1/2$ and scale 1.

Problem 10 C&B Exercise 3.24, p. 131

Many "named" distributions are special cases of the more common distributions already discussed. For each of the following named distributions derive the form of the pdf, verify that it is a pdf, and calculate the mean and variance.

(a) If $X \sim$ exponential(β), then $Y = X^{1/\gamma}$ has the *Weibull*(γ, β) distribution, where $\gamma > 0$ is a constant.

(b) If $X \sim$ exponential(β), then $Y = (2X/\beta)^{1/2}$ has the *Rayleigh distribution*.

(c) If $X \sim$ gamma(a, b), then $Y = 1/X$ has the *inverted gamma* IG(a, b) distribution. (This distribution is useful in Bayesian estimation of variances; see Exercise 7.23.)

(d) If $X \sim$ gamma$(\frac{3}{2}, \beta)$, then $Y = (X/\beta)^{1/2}$ has the *Maxwell distribution*.

(e) If $X \sim$ exponential(1), then $Y = \alpha - \gamma \log X$ has the *Gumbel*(α, γ) distribution, where $-\infty < \alpha < \infty$ and $\gamma > 0$. (The Gumbel distribution is also known as the *extreme value distribution*.)

Solution

Throughout this problem, Gamma(a, b) uses shape $a > 0$ and scale $b > 0$:

$$f_X(x) = \frac{x^{a-1} e^{-x/b}}{\Gamma(a) b^a}, \qquad x > 0.$$

In particular, Exponential(β) has mean β and density $\beta^{-1} e^{-x/\beta}$. For a one-to-one transformation $y = g(x)$, we use $f_Y(y) = f_X(g^{-1}(y))|(g^{-1})'(y)|$ on the transformed support.

(a) Weibull: $X \sim \text{Exponential}(\beta)$ and $Y = X^{1/\gamma}$, where $\gamma > 0$.

Here $x = y^\gamma$, $dx/dy = \gamma y^{\gamma-1}$, and $y > 0$. Therefore

$$\boxed{f_Y(y) = \frac{\gamma}{\beta} y^{\gamma-1} e^{-y^\gamma/\beta}, \qquad y > 0,}$$

with density zero otherwise. This is the book's Weibull(γ, β) parameterization. It is nonnegative, and $u = y^\gamma/\beta$ gives

$$\int_0^\infty f_Y(y)\, dy = \int_0^\infty e^{-u}\, du = 1.$$

The same substitution gives, for $m > -\gamma$,

$$\mathbb{E}Y^m = \beta^{m/\gamma} \int_0^\infty u^{m/\gamma} e^{-u}\, du = \beta^{m/\gamma} \Gamma\left(1 + \frac{m}{\gamma}\right).$$

Taking $m = 1, 2$,

$$\boxed{\mathbb{E}Y = \beta^{1/\gamma} \Gamma\left(1 + \frac{1}{\gamma}\right),}$$

$$\boxed{\operatorname{Var} Y = \beta^{2/\gamma} \left[\Gamma\left(1 + \frac{2}{\gamma}\right) - \Gamma\left(1 + \frac{1}{\gamma}\right)^2\right].}$$

(b) Rayleigh: $X \sim \text{Exponential}(\beta)$ and $Y = (2X/\beta)^{1/2}$.

The inverse is $x = \beta y^2/2$ and $dx/dy = \beta y$, so

$$\boxed{f_Y(y) = y e^{-y^2/2}, \qquad y > 0,}$$

and $f_Y(y) = 0$ otherwise. This density does not depend on β. Using $u = y^2/2$,

$$\int_0^\infty y e^{-y^2/2}\, dy = \int_0^\infty e^{-u}\, du = 1,$$

and, for $m > -2$,

$$\mathbb{E}Y^m = 2^{m/2} \Gamma\left(1 + \frac{m}{2}\right).$$

Since $\Gamma(3/2) = \sqrt{\pi}/2$, we obtain

$$\boxed{\mathbb{E}Y = \sqrt{\frac{\pi}{2}}, \qquad \operatorname{Var} Y = 2 - \frac{\pi}{2} = \frac{4 - \pi}{2}.}$$

(c) Inverted gamma: $X \sim \text{Gamma}(a, b)$ and $Y = 1/X$.

Here $x = 1/y$ and $|dx/dy| = y^{-2}$. Thus

$$\boxed{f_Y(y) = \frac{1}{\Gamma(a) b^a}\, y^{-a-1} e^{-1/(by)}, \qquad y > 0,}$$

with density zero otherwise. Nonnegativity is immediate. With $u = 1/(by)$, reversing the integration limits,

$$\int_0^\infty f_Y(y)\, dy = \frac{1}{\Gamma(a)} \int_0^\infty u^{a-1} e^{-u}\, du = 1.$$

More generally, for $m < a$,

$$\mathbb{E}Y^m = \mathbb{E}X^{-m} = b^{-m} \frac{\Gamma(a-m)}{\Gamma(a)}.$$

The integral diverges when $m \geq a$ for positive m. Consequently,

$$\boxed{\mathbb{E}Y = \frac{1}{b(a-1)}, \qquad a > 1,}$$

and, for $a > 2$,

$$\begin{aligned} \text{Var}\, Y &= \frac{1}{b^2(a-1)(a-2)} - \frac{1}{b^2(a-1)^2} \\ &= \boxed{\frac{1}{b^2(a-1)^2(a-2)}}. \end{aligned}$$

For $0 < a \leq 1$, the mean is infinite and the variance is undefined; for $1 < a \leq 2$, the mean is finite but the variance is infinite.

(d) Maxwell: $X \sim \text{Gamma}(3/2, \beta)$ and $Y = (X/\beta)^{1/2}$.

The inverse is $x = \beta y^2$, with $dx/dy = 2\beta y$. Hence

$$f_Y(y) = \frac{(\beta y^2)^{1/2} e^{-y^2}}{\Gamma(3/2)\beta^{3/2}} (2\beta y) = \boxed{\frac{4}{\sqrt{\pi}} y^2 e^{-y^2}}, \qquad y > 0.$$

The density is zero otherwise and does not depend on β. With $u = y^2$,

$$\int_0^\infty f_Y(y)\, dy = \frac{2}{\sqrt{\pi}} \int_0^\infty u^{1/2} e^{-u}\, du = \frac{2}{\sqrt{\pi}} \Gamma(3/2) = 1.$$

The moments are

$$\mathbb{E}Y^m = \frac{\Gamma((m+3)/2)}{\Gamma(3/2)}, \qquad m > -3.$$

Therefore

$$\boxed{\mathbb{E}Y = \frac{2}{\sqrt{\pi}}, \qquad \text{Var}\, Y = \frac{3}{2} - \frac{4}{\pi}.}$$

(e) Gumbel: $X \sim \text{Exponential}(1)$ **and** $Y = \alpha - \gamma \log X$**, where** $\alpha \in \mathbb{R}$ **and** $\gamma > 0$**.**
Put $z = (y-\alpha)/\gamma$. The inverse is $x = e^{-z}$ and $|dx/dy| = \gamma^{-1}e^{-z}$. Thus

$$\boxed{f_Y(y) = \frac{1}{\gamma} \exp\left\{-\frac{y-\alpha}{\gamma} - \exp\left(-\frac{y-\alpha}{\gamma}\right)\right\}, \qquad -\infty < y < \infty.}$$

This density is nonnegative. Using $u = e^{-(y-\alpha)/\gamma}$,

$$\int_{-\infty}^{\infty} f_Y(y)\, dy = \int_{\infty}^{0} -e^{-u}\, du = 1.$$

To compute the moments, for $t < 1/\gamma$ the gamma integral gives

$$M_Y(t) = \mathbb{E}(e^{tY}) = e^{\alpha t}\mathbb{E}(X^{-\gamma t}) = e^{\alpha t}\Gamma(1-\gamma t).$$

Let γ_E denote Euler's constant, to distinguish it from the scale parameter γ. The gamma-function identities needed here are

$$\frac{d}{du}\log\Gamma(u)\bigg|_{u=1} = -\gamma_E, \qquad \frac{d^2}{du^2}\log\Gamma(u)\bigg|_{u=1} = \sum_{k=1}^{\infty}\frac{1}{k^2} = \frac{\pi^2}{6}.$$

For example, both follow by differentiating Euler's product

$$\Gamma(1+s) = e^{-\gamma_E s}\prod_{k=1}^{\infty}\left(1+\frac{s}{k}\right)^{-1} e^{s/k}$$

near $s = 0$. Differentiating the cumulant generating function $K_Y(t) = \log M_Y(t) = \alpha t + \log\Gamma(1-\gamma t)$ at zero now yields

$$\boxed{\mathbb{E}Y = K_Y'(0) = \alpha + \gamma\gamma_E, \qquad \operatorname{Var} Y = K_Y''(0) = \frac{\pi^2\gamma^2}{6}.}$$